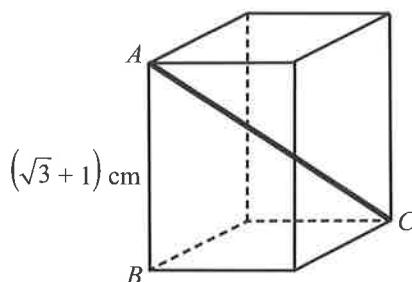


Basic

1. A rectangle has length $(3 + 2\sqrt{3})$ cm and an area of $(6 + 5\sqrt{3})$ cm². Find, **without using a calculator**, the width of the rectangle, in cm, in the form $(a + b\sqrt{3})$, where a and b are integers. [3]
[N21/I/1]
2. A rectangular block has a square base of side $(\sqrt{6} + \sqrt{2})$ cm and a height of h cm. The volume of the rectangular block is $(16 + 4\sqrt{3})$ cm³. **Without using a calculator**, show that h can be expressed as $a + b\sqrt{3}$, where a and b are integers. [5]
[N16/I/2]

Intermediate

3. A circular cylinder of volume $(3\sqrt{7} - 6)\pi$ cm³ has a height of $(2 + \sqrt{7})$ cm and a radius of r cm. Without using a calculator, obtain an expression for r^2 in the form $(a + b\sqrt{7})$, where a and b are integers. [4]
[N19/I/10(b)]
4. (a) Given that $3 + 2\sqrt{5}$ is a root of the equation $x^2 + ax + b = 0$, where a and b are integers, find the value of a and of b . [3]
(b) A rectangle of length $(6 + \sqrt{12})$ cm has an area of $(24 + \sqrt{48})$ cm². **Without using a calculator**, find the breadth of the rectangle, in cm, in the form $(c + d\sqrt{3})$, where c and d are integers. [3]
[N18/I/4]
5. The triangle ABC is such that its area is $\frac{1}{4}(9 + \sqrt{3})$ cm², the length of AB is $(\sqrt{3} + 1)$ cm and angle BAC is 60° . **Without using a calculator**, find
(i) the length, in cm, of AC in the form $a + b\sqrt{3}$, where a and b are integers, [3]
(ii) an expression, in cm², for BC^2 in the form $c + d\sqrt{3}$, where c and d are integers. [3]
[N17/I/7]
6. **Do not use a calculator in this question.**
(i) Express $\frac{11\sqrt{3}}{2\sqrt{3} + 1}$ in the form $a + b\sqrt{3}$, where a and b are integers. [2]



The diagram shows a cuboid with a square base. The height AB of the cuboid is $(\sqrt{3} + 1)$ cm.

Given that the length of the diagonal AC is $\frac{11\sqrt{3}}{2\sqrt{3} + 1}$ cm,

- find an expression for BC^2 in the form $c + d\sqrt{3}$, where c and d are integers, [3]
- express the volume of the cuboid in the form $\frac{7}{2}(3\sqrt{3} + k)$ cm³, where k is an integer. [4]

[N15/III/5]

